2025 Spring

II.

($\Leftarrow$) $r(A+B) = r(A) + r(B)$

$\Leftrightarrow C(A+B) = C(A) \oplus C(B)$, $C(A) \cap C(B) = \{0\}$.

$(A+B)(A+B)^-(A+B) = A+B$

$\Rightarrow A(A+B)^-A + A(A+B)^-B + B(A+B)^-A + B(A+B)^-B$.

$= A+B$

Here, $\underset{\in C(A)}{A} = CC^-A = \underset{\in C(A)}{AC^-A} + \underset{\in C(B)}{BC^-A}$

$\text{rank}(AC^-A) \le \text{rank}(A)$

$\Rightarrow C(AC^-A) \subset C(A)$

& $C(BC^-A) \subset C(B)$

$\Rightarrow AC^-A = A$, $BC^-A = 0$ $\because C(A) \oplus C(B) = \{0\}$.

Similarly, $\underset{\in C(B)}{B} = CC^-B = \underset{\in C(A)}{AC^-B} + \underset{\in C(B)}{BC^-B}$

$\Rightarrow AC^-B = 0$, $BC^-B = B$ $\because C(A) \oplus C(B) = \{0\}$.

($\Rightarrow$) $A = (A+B)C^-A = CC^-A$

$B = CC^-B$.

Let $x \in C(A) \cap C(B)$, $\exists u, v$ s.t. $x = Au = Bv$.

$\Rightarrow x = AC^-x$ $\because x \in C(A)$, $AC^-A = A$

$\& \ x = Bv.$

$\Rightarrow AC^{-}x = AC^{-}Bv = 0$

$\Rightarrow x = 0$

$\Rightarrow C(A) \cap C(B) = \{0\}$

$\Rightarrow C(C) = C(A) \oplus C(B)$

$\Rightarrow r(C) = r(A) + r(B).$ $\blacksquare$

$\boxed{2}$.

$$\frac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{n^{-1}}} \sim N(0,1)$$

$$\frac{\hat{n} S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \chi^2(n_1-1) + \chi^2(n_2-1)$$
$$\sim \chi^2(n_1+n_2-2)$$

$$\Rightarrow \frac{\hat{n} S_p^2}{\sigma^2} \Big/ \hat{n} \sim \chi^2(n_1+n_2-2) \Big/ \hat{n}$$

$$\bar{y}_1, \bar{y}_2 \perp\!\!\!\perp S_1^2, S_2^2 \Longrightarrow \bar{y}_1 - \bar{y}_2 \perp\!\!\!\perp S_p^2$$

$$t = \frac{\dfrac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{n^{-1}}}}{\sqrt{\dfrac{\hat{n} S_p^2}{\sigma^2} \Big/ \hat{n}}} = \frac{Z}{\sqrt{\chi^2(\hat{n})/\hat{n}}}$$
$$\sim t(n_1+n_2-2)$$

$$t^2 \sim F(1, \hat{n})$$

$\blacksquare$

$\boxed{3}$.

($\Leftarrow$) $X^TX(\hat{\beta} + (I - X^-X)z)$

$= X^TX\hat{\beta} + (X^TX - \underbrace{X^TX X^-X}_{=X})z$

$= X^Ty$: solution.

($\Rightarrow$) Let $\tilde{\beta}$ be a solution of $X^TX\beta = X^Ty$

We need to show that $\exists z \in \mathbb{R}^p$ s.t. $\tilde{\beta} = \hat{\beta} + (I - X^-X)z$

Put $z = \tilde{\beta} - \hat{\beta}$.

$\hat{\beta} + (I - X^-X)(\tilde{\beta} - \hat{\beta})$

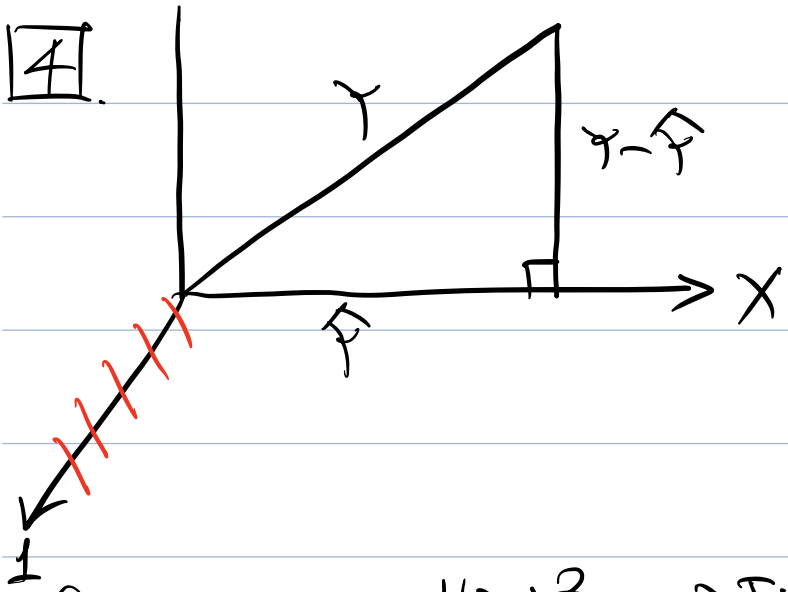
$= \hat{\beta} + \tilde{\beta} - \hat{\beta} - X^-X\tilde{\beta} - X^-X\hat{\beta}$

$= \tilde{\beta} - X^Ty + X^Ty$

$= \tilde{\beta}$

$\blacksquare$

4.



(a) $SST = \|Y\|^2 = Y^T Y$

$SSE = \|Y - \hat{Y}\|^2 = Y^T (I - H_X) Y$

$SSR = \|\hat{Y}\|^2 = Y^T H_X Y$

$\because$ check : $SST = SSE + SSR$

$$Y^T Y = Y^T (I - H_X) Y + Y^T H_X Y$$

(b) $H_0 : \beta_1 = \cdots = \beta_p = 0$ vs. H_1 : not H_0

$$F = \frac{SSR / p}{SSE / n-p} = \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / n-p}$$

① $\dfrac{Y^T H_X Y}{\sigma^2} \sim \chi^2\left(p, \dfrac{\beta^T X^T X \beta}{2\sigma^2}\right)$

$\text{rank}(\underbrace{H_X}_{\text{idemp}}) = tr(H_X) = p$, $\frac{1}{2} \frac{\mu^T H_X \mu}{\sigma} = \dfrac{\beta^T X^T X \beta}{2\sigma^2}$

② $\dfrac{Y^T(I-H_X)Y}{\sigma^2} \sim \chi^2(n-p)$

$$\text{rank}(\underbrace{I-H_X}_{\text{idemp}}) = tr(I-H_X) = n-p$$

$$\frac{1}{2}\frac{\mu}{\sigma}(I-H_X)\frac{\mu}{\sigma} = \frac{1}{2}\frac{\mu}{\sigma}(X-X)\frac{\beta}{\sigma} = 0$$

③ Independence $\quad H_X(I-H_X) = H_X - H_X = 0$

$$F = \frac{SSR/p}{SSE/n} \begin{matrix} \overset{H_1}{\sim} F\left(p, n-p, \frac{\beta^T X^T X \beta}{2\sigma^2}\right) \\ \overset{H_0}{\sim} F(p, n-p) \end{matrix}$$

$\blacksquare$

$\boxed{5}$.

$$\hat{\varepsilon}^* = (I - H_{X_1})Y$$

$$\hat{\varepsilon}^* = X_2\beta_2^* + \gamma$$

$$\hat{\eta} = (I - H_{X_2})\hat{\varepsilon}^*$$

$$= (I - H_{X_2})(I - H_{X_1})Y$$

$$\frac{\hat{\eta}^T\hat{\eta}}{\sigma^2} = \frac{Y^T(I - H_{X_1})(I - H_{X_2})(I - H_{X_2})(I - H_{X_1})Y}{\sigma^2}$$

$$= \frac{Y^T(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})Y}{\sigma^2}$$

$(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})$ should be Idempotent.

$$(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})$$

$$= (I - H_{X_1})(I - H_{X_2})(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})$$

should be $(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})$

If $X_1^T X_2 = 0$,

$$(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})(I - H_{X_2})(I - H_{X_1})$$

$$= (I - H_{X_1} - H_{X_2})(I - H_{X_1})(I - H_{X_1} - H_{X_2})$$

$$= (I - H_{X_1} - H_{X_2})(I - H_{X_1} - H_{X_2})$$

$$= I - H_{X_1} - H_{X_2}$$

& $(I - H_{X_1})(I - H_{X_2})(I - H_{X_1}) = I - H_{X_1} - H_{X_2}$

so, $(I-H_{X_1})(I-H_{X_2})(I-H_{X_1}) = I-H_{X_1}-H_{X_2}$

is idempotent under $X_1^T X_2 = 0$

$tr(I-H_{X_1}-H_{X_2}) = n-p-q$

$(\beta_1^T X_1^T + \beta_2^T X_2^T)(I-H_{X_1}-H_{X_2})(X_1\beta_1 + X_2\beta_2)$

$= 0$

so, $\frac{\hat{\eta}^T\hat{\eta}}{\sigma^2} \sim \chi^2(n-p-q)$

$\boxtimes$

$\boxed{6}$.

(a) $U = (y - X\beta)^T (y - X\beta) + \lambda^T (C\beta - 0)$

$$\frac{\partial U}{\partial \lambda} = C\beta = 0$$

$$\frac{\partial U}{\partial \beta} = -2X^T y - 2X^T X\beta + C^T\lambda = 0$$

$$\Rightarrow \hat{\beta}_c = (X^TX)^{-1}X^Ty - \frac{1}{2}(X^TX)^{-1}C^T\lambda$$

Since $C\hat{\beta}_c = 0$,

$$C(X^TX)^{-1}X^Ty - \frac{1}{2}C(X^TX)^{-1}C^T\lambda = 0$$

$$\lambda = 2[C(X^TX)^{-1}C^T]^{-1}C(X^TX)^{-1}X^Ty$$

$$\hat{\beta}_c = \hat{\beta} - \frac{1}{2}2\cdot(X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}C(X^TX)^{-1}X^Ty$$

$$= \hat{\beta} - (X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}C\hat{\beta}$$

$\boxtimes$

(b) $B := (X^TX)^{-1}C^T(C(X^TX)^{-1}C^T)^{-1}C$: idempotent

$I - B$: idempotent.

$Qx = 0 \Rightarrow x = 0$

$$\begin{pmatrix} C \\ \cdots \\ P_1^T \end{pmatrix} \begin{matrix} q \\ \\ p+1-q \end{matrix}$$

$tr(I - B) = p+1-q$

Let $Qx = 0 \Rightarrow Cx = 0$ & $P_1^Tx = 0$

If $P_1^Tx = 0 \Leftrightarrow x \in N(P_1^T) \Leftrightarrow x \in C(P_2)$

$\Rightarrow x = P_2 v$ for some v

$\text{rank}(P_1^T) = p+1-q$

$(I-B)P_2 = 0 \Rightarrow (I-B)P_2 v = (I-B)d = 0$

$d - Bd = 0 \Rightarrow d = Bd$

$\Rightarrow d = (X^TX)^{-1}C^T(C(X^TX)^{-1}C^T)^{-1}\underbrace{Cd}_{=0}$

$= 0$

$\boxtimes$